



ANDERSON JUNIOR COLLEGE
2013 JC 2 PRELIMINARY EXAMINATIONS

CHEMISTRY

9647/03

Higher 2

19 September 2013

Paper 3 Free Response Questions

2 hours

Candidates answer on separate paper.

Additional Materials: Data Booklet
 Writing paper

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.

Start each question on a fresh sheet of paper.

A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This document consists of **12** printed pages.

- 1 (a) Nicotinamide adenine dinucleotide (NAD^+) is a coenzyme responsible for carrying electrons from one reaction to another in living cells.

The electrode potential for the reduction of NAD^+ in a biological system, $E(\text{pH } 7)$, in which the conditions are at 1 mol dm^{-3} , 25°C and $\text{pH } 7$, is as shown. The oxidised and reduced form of the coenzyme is represented as NAD^+ and NADH respectively.



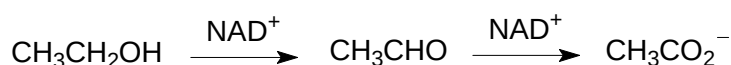
The following Nernst equation can be used to calculate the electrode potential of the above system when $[\text{NAD}^+]$ and $[\text{NADH}]$ change.

$$E = E(\text{pH } 7) - \frac{0.0592}{n} \log_{10} \frac{[\text{NADH}]}{[\text{NAD}^+]}$$

where n is the number of moles of electrons transferred in the system.

- (i) Using the Nernst equation given, determine the ratio $\frac{[\text{NADH}]}{[\text{NAD}^+]}$ when $E = -0.350 \text{ V}$.
- (ii) Hence, calculate the percentage of NAD^+ in the $\text{NAD}^+ - \text{NADH}$ mixture when $E = -0.350 \text{ V}$.

NAD^+ is used to metabolise ethanol in living cells via a series of reactions. In the first stage, ethanol is oxidised to ethanal. In the second stage, ethanal is converted to ethanoate ions.



The electrode potential, $E(\text{pH } 7)$, and Gibbs free energy change, $\Delta G(\text{pH } 7)$, of each of the carbon-containing species present in the two stages of the metabolism are given below.

Half-equation	$E(\text{pH } 7) / \text{V}$	$\Delta G(\text{pH } 7) / \text{kJ mol}^{-1}$
$\text{CH}_3\text{CHO} + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{CH}_3\text{CH}_2\text{OH}$	-0.197	+38
$\text{CH}_3\text{COO}^- + 3\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{CH}_3\text{CHO} + \text{H}_2\text{O}$	-0.581	+112

- (iii) Using the data given above, calculate the $E_{\text{cell}}(\text{pH } 7)$ for **each** of the two stages of the metabolism of ethanol.
- (iv) Write a half-equation for the conversion of ethanoate ion to ethanol.

Hence, using your knowledge of Hess' Law for ΔG , calculate $\Delta G(\text{pH } 7)$ for this conversion.

(v) $\Delta G(\text{pH } 7)$ and $E(\text{pH } 7)$ are related by the following equation.

$$\Delta G(\text{pH } 7) = -zFE(\text{pH } 7)$$

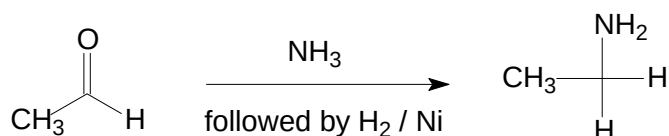
where $\Delta G(\text{pH } 7)$ is the Gibbs free energy change in **joules per mole**, z is the number of moles of electrons transferred during the redox reaction and F is the Faraday constant.

Use the equation you have written in (a)(iv) to decide on a suitable value of z , and hence calculate the value of $E(\text{pH } 7)$.

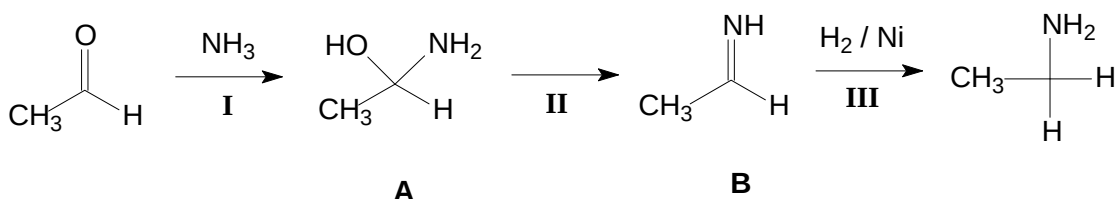
[9]

(b) Reductive amination is a common method used in the pharmaceutical industry that converts carbonyl compounds to amines.

One example of this method is the preparation of ethylamine. Ethanal can be converted to ethylamine by reductive amination with ammonia, using hydrogen gas over a nickel catalyst as the reducing agent.



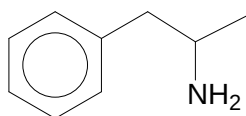
The reaction is thought to proceed through the following steps.



- (i) Given that ammonia acts as a nucleophile in step I, suggest a mechanism for this step. In your answer show any relevant charges, dipoles or lone pairs of electrons you consider important in this mechanism.
- (ii) State the type of reaction for step II.
- (iii) Suggest, in terms of the mechanism you have described in (b)(i), whether compound A has any effect on plane-polarised light.

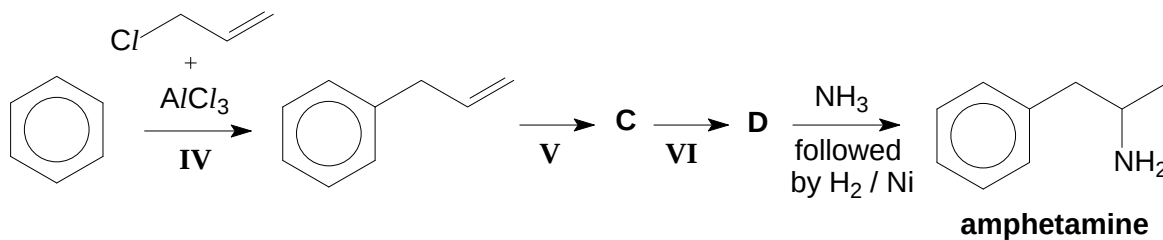
[6]

- (c) Amphetamine is a synthetic psychoactive drug that stimulates the central nervous system by increasing the heart rate and blood pressure.



amphetamine

Reductive amination shown in (b) can be used as the last step in the synthesis of amphetamine from benzene.



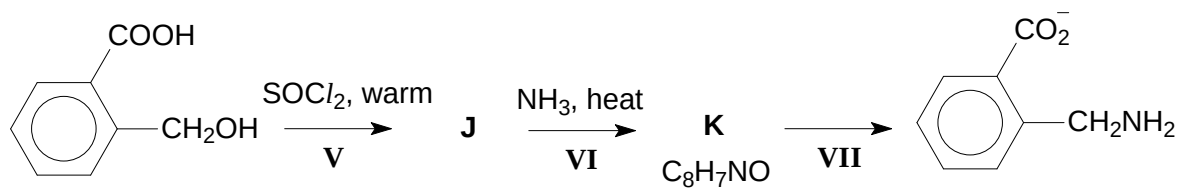
- (i) State the type of reaction for step **IV**.
- (ii) Suggest the reagents and conditions for steps **V** and **VI** and the structures for **C** and **D**.

[5]
[Total: 20]

- [8]

- [5]

(c) 2-(hydroxymethyl)benzoic acid undergoes the following reaction scheme.



- (i) Suggest the structural formulae of **J** and **K**.
- (ii) Step **VII** takes place in the presence of a base. Name the type of reaction that has taken place.

The thionyl chloride, SOCl_2 used in step **V** can be formed from the reaction of SO_2 with PCl_5 . A byproduct **L** with $M_r = 153.5$ is also formed.

- (iii) Explain, with the aid of an equation, why the synthesis of SOCl_2 has to be carried out under anhydrous conditions.
- (iv) Suggest a formula for **L** and construct an equation for its production from the reaction of SO_2 with PCl_5 .

[7]
[Total: 20]

- 3 (a) An experiment was conducted to investigate which mechanism, S_N1 or S_N2 , is involved in the hydrolysis of 2-iodobutane.

When an optically pure enantiomer of 2-iodobutane was reacted with hot, aqueous sodium hydroxide, the purified product obtained was found to rotate plane-polarised light clockwise by 5.5° .

An optically pure enantiomer of butan-2-ol was found to rotate plane-polarised light clockwise by 13.5° .

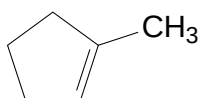
- (i) Explain the difference in optical activity shown by products formed when 2-iodobutane undergoes S_N1 and S_N2 mechanisms respectively.
- (ii) For a secondary halogenoalkane like 2-iodobutane, both S_N1 and S_N2 can occur to some extent.

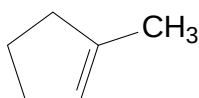
Using the information given above, determine the percentage of 2-iodobutane that underwent the S_N2 mechanism.

- (iii) Describe and explain how the rate of hydrolysis will change if 2-chlorobutane is used instead of 2-iodobutane.

[4]

(b)



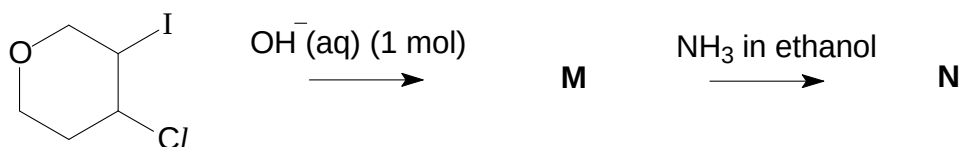
When 1-methylcyclopentene, , is bubbled through iodine monochloride, ICl , dissolved in a suitable solvent, a dihalogeno compound is formed.

Name the type of reaction and describe the mechanism for this reaction, including curly arrows showing the movement of electrons, and all charges.

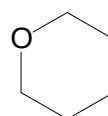
[3]

- (c) The different reactivities of organic halogeno compounds can be exploited in reactions of dihalogeno compounds.

Using your answers in (a)(iii), predict the outcomes of the following transformations, drawing the structures of the intermediate **M** and the product **N**.



In the above reactions, you may assume that the



ring remains unaltered.

[2]

- (d) Compound **P** is an isomer of compound **M**. Compound **P** is sparingly soluble in water but readily soluble in aqueous sodium hydroxide.

Heating compound **P** in aqueous sodium hydroxide, followed by adding excess acid to the product yields compound **Q**, $C_5H_{10}O_3$, which reacts with warm alkaline aqueous iodine to give a yellow precipitate.

On treatment with hot concentrated sulfuric acid, compound **Q** forms a mixture of three isomeric compounds **R**, **S** and **T**, with the formula $C_5H_8O_2$, **two** of which are geometrical isomers of each other.

Compound **R** is a sweet smelling liquid. Compound **S** reacts with 1 mole of aqueous bromine to give a number of products. Both **S** and **T** undergo reaction with acidified potassium manganate(VII) to give **U**, $C_3H_4O_4$, and ethanoic acid. 1 mole of **U** liberates 1 mole of CO_2 on reaction with excess sodium carbonate.

- (i) Identify the five compounds **Q** – **U**, explaining the chemistry of the reactions described.
- (ii) Write a balanced equation for the reaction of compound **Q** with warm alkaline aqueous iodine.

[11]
[Total: 20]

- 4 The name of chromium is derived from the Greek word 'chrōma', which means colour, because many of its compounds are colored.

- (a) When chromium metal is dissolved in dilute sulfuric acid, it produces a blue solution containing $\text{Cr}^{2+}(\text{aq})$. The blue solution slowly turns green even in the absence of air.

This mixture is then filtered to separate the green solution and the unreacted metal. The green filtrate, **V**, is divided into two portions, each of 10.0 cm^3 .

To one portion of solution **V**, adding sodium hydroxide solution produces a grey-green precipitate, **W**. In excess sodium hydroxide, a dark green solution, **X**, is formed. The dark green solution turns into a yellow solution, **Y**, upon warming with hydrogen peroxide solution.

- (i) By quoting and using relevant E° values from the *Data Booklet*, explain why the blue solution turns green.
- (ii) Suggest the formula of the complex ion present in solution **V**.
- (iii) Explain why solution **V** is green in colour.
- (iv) Identify **W** and **X** and suggest balanced equations for the formation of **W** from **V**, and for the formation of **X** from **W**.
- (v) What type of reaction occurs when **X** is converted into **Y**?

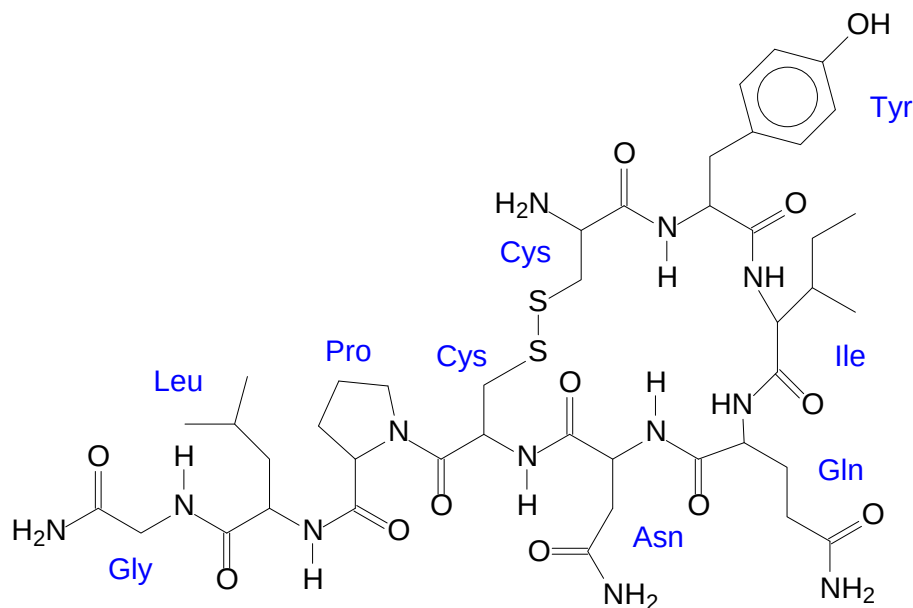
[9]

- (b) The other portion of solution **V** is reacted **exactly** with 11.0 cm^3 of $0.030 \text{ mol dm}^{-3}$ sodium carbonate, producing the same grey-green precipitate, **W**. No carbon dioxide is given off during the reaction.

- (i) Given that the pH of solution **V** is 3.0, calculate the concentration of the ion you have suggested in (a)(ii).
[K_a of the ion suggested in (a)(ii) = $9.8 \times 10^{-5} \text{ mol dm}^{-3}$]
- (ii) Using your answer to (b)(i) and the information given above, calculate the number of moles of sodium carbonate that would react with one mole of the ion in solution **V**.
- (iii) Hence, determine the ionic equation for the above reaction between sodium carbonate and solution **V**.

[4]

- (c) Oxytocin is a hormone that is secreted by the pituitary gland, and functions by stimulating uterine contraction and lactation during childbirth. The structure of the oxytocin molecule is shown below.



amino acid	formula of side chain (R in $RCH(NH_2)CO_2H$)
Gly	-H
Leu	$-CH_2CH(CH_3)_2$
Pro	$-CH_2CH_2CH_2-$
Cys	$-CH_2SH$
Asn	$-CH_2CONH_2$
Gln	$-CH_2CH_2CONH_2$
Ile	$-CH(CH_3)CH_2CH_3$
Tyr	$-CH_2C_6H_4OH$

- (i) Explain what is meant by the term denaturation?
- (ii) Suggest how chromium metal can interact with the oxytocin molecule to bring about denaturation.
- (iii) Besides chromium metal, discuss one other factor that could cause the oxytocin molecule to denature. You should use suitable amino acids listed above to illustrate your answer.
- (iv) What reagents and conditions are needed to hydrolyse the oxytocin molecule into its constituent amino acids?
- (v) Draw the displayed formula of the amino acid present in the oxytocin molecule that contains two chiral atoms. Indicate **each** chiral atom with an asterisk (*).

[7]

[Total: 20]

- 5 (a) Phosphorus sulfide, P_4S_3 , is used in small amounts in the tip of a match stick. On striking a match stick, this compound burns to form sulfur dioxide and a solid oxide.

(i) Construct an equation for this reaction.

- (ii) The melting points of the two oxides formed in (a)(i) differ significantly from that of silicon(IV) oxide (SiO_2).

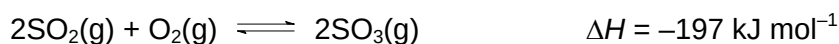
Briefly relate this observation to the structure of, and bonding in, each of the oxides.

- (iii) The solid oxide formed in (a)(i) dissolves in water to give an acidic solution.

Construct an equation for this reaction and give an approximate pH for the solution formed.

[6]

- (b) The Contact Process is an industrial process for manufacturing sulfuric acid. The key stage in this process is the reaction between sulfur dioxide and oxygen.



- (i) Write an expression for the equilibrium constant, K_p , giving its units.

This reaction is carried out at a pressure of 5 atm and a temperature of 500 °C.

- (ii) In what respects are these conditions **not** what you might have expected? Explain your answer.

- (iii) Vanadium(V) oxide is used as a *heterogeneous catalyst* for this process.

Suggest why vanadium(V) oxide can act as a *heterogeneous catalyst*.

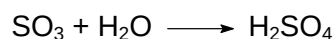
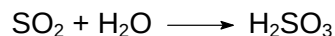
- (iv) The sulfur trioxide that is produced is then converted into concentrated sulfuric acid. Concentrated form of the acidic solutions, formed here and in (a)(iii), differ in their reactions with sodium iodide.

Write balanced equations to illustrate this difference.

[8]

- (c) In an experiment, a mixture of 0.2 mol of SO_2 and 0.1 mol of O_2 was allowed to reach equilibrium at 500°C and a total pressure of 5 atm.

The gases were then rapidly cooled and an excess of water was very carefully added to the mixture of gases, under suitable conditions, causing the following reactions to occur.



The resulting solution was made up to 250 cm^3 in a standard flask. 25.0 cm^3 of this solution was titrated with 1.00 mol dm^{-3} NaOH and required 39.20 cm^3 for neutralisation.

You may assume that H_2SO_3 behaves as a monobasic acid in the determination of end point in the titration.

- (i) Calculate the total amount, in moles, of H^+ ions present in the 250 cm^3 standard flask.
- (ii) Hence, calculate the amounts, in moles, of SO_2 , O_2 and SO_3 in the original equilibrium at 500°C .
- (iii) Use your answers to (b)(i) and (c)(ii) to calculate a value for K_p for the equilibrium.

[6]

[Total: 20]